

Motor Datasheet: WxF-85x28GT

Machine Configuration

Parameter	Value	Unit
Machine Outer Diameter	83.0	mm
Machine Length / Thickness	28.1	mm
Voltage	48.0	V
Operating Temperature	-40 to 85	°C

Mass Properties

Parameter	Value	Unit
Machine Mass (active components only)	752.0	g
Rotor Mass	255.3	g
Stator Mass	496.7	g
Rotor Structure Included	No	-
Rotor Moment of Inertia	2497.4	g·cm ²

Performance Metrics

Parameter	Value	Unit
Torque (continuous)	5.40	Nm
Speed	2500	rpm
Losses (continuous)	98.8	W
Supply Current (RMS)	26.99	A
Phase Resistance (per phase)	35.0	mΩ
Terminal Inductance ($L_{tt} = L_d + L_q$)	238.2	μH
D-axis Inductance (L_d)	121.3	μH
K_e (peak, phase)	9.65	mV/rpm
K_e (peak, line-to-line)	16.71	mV/rpm
K_t (peak)	137.1	mNm/A
Motor Constant (K_m)	616.5	mNm/√W